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Effect of hydrogen on the ideal shear strength in metals and its implications on plasticity: A first-principles study

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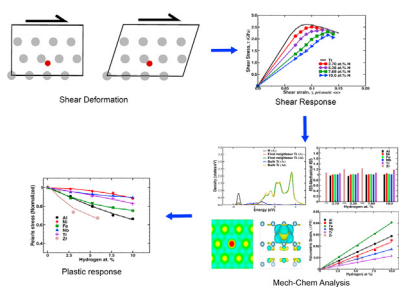
HIGHLIGHTS

- The volumetric strain by hydrogen had a key role in the shear strength response.
- The uninterrupted charge depletion near hydrogen led to a noticeable drop in strength.
- The Peierls stress was found to decrease with increasing hydrogen concentration.

GRAPHICAL ABSTRACT



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ABSTRACT

Hydrogen embrittlement limits the service life of various metallic components by causing a transition from a ductile to a brittle failure of inherently ductile alloys. In this work, using first-principles calculations, the effect of interstitial hydrogen on the ideal shear strength across various metals (Al, Ni, Fe, Nb, Ti, and Zr) and its implications on plasticity are discussed. The presence of hydrogen led to a volumetric expansion, which in turn had a key role in the observed shear strength response of cubic metals. However, in the case of HCP metals, the chemical contributions also have a significant part in the observed shear strength response. The interstitial hydrogen atom interacts strongly with valence *d* orbital metals (Ni, Fe, Nb, Ti, and Zr). Based on the Peierls-Nabarro framework, the presence of interstitial hydrogen reduces the Peierls stress across all the metals examined here. Finally, these findings provide insights to comprehensively understand hydrogen embrittlement.

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Introduction

The adsorption and subsequent permeation of atomic hydrogen into intrinsically ductile metallic alloys leads to drastic degradation of mechanical properties, this phenomenon is known as hydrogen embrittlement (HE) [1,2]. Over the past few decades, HE has attracted significant research interest across various metallic alloys [1–8]. For instance, in Al and Ni alloys, the adequate accumulation of hydrogen along the grain boundaries (GBs) can lead to the formation of intergranular cracks [9–11]. These microscopic events cause a noticeable decrease in fracture strength, tensile strength, and ductility [10–13]. In Al alloys, the interstitial hydrogen was found to be bind with multiple vacancies, which causes hydrostatic strain localization leading to macroscopic fracture [12–14]. Interestingly in Ni alloys, the presence of hydrogen atoms that were not bound to energetically favorable defect (dislocations, GBs, and interface of second phase particles) had a limited role in the overall deformation behavior [10]. Similarly, the effect of hydrogen on the mechanical behavior of various ferrous alloys has attracted a great deal of attention [1,2,15–17]. Broadly, the presence of hydrogen induces a "quasi-cleavage" fracture, where the crack propagation happens along the non-cleavage planes [1]. However, plasticity has an important role in ferrous alloys by enabling: a) the formation of strain localization due to dislocation pileups at the GBs; and b) transportation of hydrogen atoms to the GBs, as in the low temperature and strain rate regime hydrogen atoms can travel along with dislocations [1]. Further, in the case of niobium, titanium, and zirconium alloys, a brittle hydride phase forms when the hydrogen concentration exceeds the solubility limit, resulting in a drastic reduction in ductility and fracture toughness [18–20]. Nonetheless, at the microscopic level hydrogen enhanced plasticity has a secondary role in the overall mechanical degradation and failure [3,21]. Overall, the severity of HE varies based on numerous factors (mode of hydrogen entry (chemical vs. gaseous), alloy composition, microstructure features, temperature, strain rate, etc.) that have been extensively reviewed [22–25].

Despite these efforts, there is a lack of consensus on the underlying mechanism(s) responsible for HE [26–30]. For instance, the hydrogen enhanced decohesion mechanism postulates that the reduction in the surface energy in the presence of hydrogen is the key factor for the ductile to brittle transition [31–35]. The hydrogen enhanced localized plasticity (HELP) mechanism is based on several experimental observations and suggests that HE occurs due to enhanced dislocation activity [1,5,6,15,36–39]. Unlike the HELP mechanism, the adsorption-induced dislocation emission suggests that the adsorption of hydrogen at crack tip and other stress concentration sites leads to the weakening of interatomic bonds, thereby facility dislocation emission at lower stress levels [40]. The dislocation emission from the crack tip leads to the build-up of high stresses in the plastic zone that eventually leads to void nucleation and coalescence with the existing crack tip. On the other hand, the defactant mechanism proposed that the presence of hydrogen leads to a significant reduction in the defect formation energy [41,42]. In pipeline steels another mechanism has been proposed to describe HE [17], here the adsorption and subsequent sub-surface trapping of hydrogen leads to the recombination of hydrogen. The continuous build-up of hydrogen at these trap sites allows for recombination, which raises the internal hydrogen pressure and eventually leads to blistering and cracking [43]. This clearly highlights the ambiguity concerning the HE mechanism, which in turn hinders the ability to effectively mitigate it.

HE is a multi-physics phenomenon that spans across several lengths and time scales, which presents a complex challenge to comprehensively understanding the underlying mechanism. This in turn limits the ability of experimental techniques to serve as the only line of inquiry to understand HE. Therefore, this has motivated a great deal of theoretical and computational research that can be synergistically combined with experimental insights [44–53]. There are several microscopic level defect-hydrogen interactions that influence the macroscopic mechanical response. Therefore, in this work, we intend to examine the upper bound of hydrogen effects on the homogenous dislocation nucleation and dislocation mobility across different metals. Towards this, the ideal shear strength (τ_{IS}), which is the stress required to plastically deform a defect-free single crystal under shear deformation was quantified. The τ_{IS} is an intrinsic material property that depends on the underlying crystalline lattice and the interatomic bonding. Thus, the quantification of the ideal shear strength through first-principles calculations provides an opportunity to examine the effect of solute atoms on the plastic deformation [54–57]. The ideal shear strength is also a key material parameter in the approximation of Peierls stress (σ_p), which is a measure of the stress required to move dislocation in the crystal lattice (in the absence of temperature) and one of the essential parameter for modeling the crystal plasticity behavior at the continuum scale [58].

$$\sigma_p = \frac{Kb}{a'} e^{\frac{-2\pi\zeta}{a'}} \tag{1}$$

where b is the magnitude of the Burger's vector and a' is the atomic spacing in the direction of the dislocation motion. The elastic parameter K can be formulated as:

$$K = \mu\left(\frac{\sin^2\theta}{1-\vartheta} + \cos^2\theta\right) \tag{2}$$

where μ is the isotropic shear modulus and ϑ is the Poisson's ratio. The variation in the dislocation character is accounted by the angle θ, where $\theta = 0^o$ and 90° for the screw and the edge dislocation, respectively. Finally, ζ is the half-width of the dislocation, which can be expressed in the following form:

$$\zeta = \frac{Kb}{4\pi\tau_{IS}} \tag{3}$$

This article aims to assess the effect of interstitial hydrogen on the ideal shear strength (τ_{IS}) and its implications on the plasticity of various metals (Al, Ni, Fe, Nb, Ti, and Zr). First, a series of first-principles calculations were carried out to quantify the variation in lattice resistance to shear deformation, due to the presence of hydrogen. Next, these findings were interpreted with the help of electronic insights using the density of states, Bader charge analysis, Electron localization function, and charge density redistribution to highlight the

subtle differences of metal-hydrogen interactions and their influence on the shear deformation. Then, the chemical and mechanical contributions of interstitial hydrogen on the ideal shear strength was separated for the various metals examined here. Finally, the implications of these findings on the dislocation mobility were discussed with the aid of the Peierls-Nabarro framework.

Methodology

To quantify the effect of hydrogen on the ideal shear strength of various metallic systems, first-principles calculations based on the density functional theory (DFT) were carried out using the Vienna Ab-initio Simulation Package (VASP) [59]. The projected augmented wave pseudo-potentials were used to simulate various metals and hydrogen atoms. Furthermore, the pseudopotential with the following valence configuration description were utilized: Al ($3s^2 3p^1$), Ni ($3d^8\ 4s^2$), Fe ($3p^6 3d^7\ 4s^1$), Nb ($4p^6 4d^7\ 5s^1$), Ti ($3d^3\ 4s^1$) and Zr ($4d^3\ 5s^1$) respectively. The exchange-correlation part of the energy was treated using the Perdew-Burke-Ernzerhof (PBE) generalized gradient approximation (GGA) [59–61] with a Methfessel-Paxton smearing width of 0.1 eV for all the material, except for Zr where the Perdew-Wang (PW91) parameters were employed. Furthermore, an energy cutoff of 400 eV, 540 eV, 600 eV, 550 eV, 520 eV, and 520 eV were selected for Al, Ni, Fe, Nb, Ti, and Zr, respectively. The calculations for Fe and Ni were spin-polarized, while the others were non-spin polarized. The Monkhorst pack and gamma centered k-point mesh was employed based on the cubic (Al, Ni, Fe, and Nb) and hexagonal (Ti and Zr) symmetry, respectively. For all calculations, we used grid width for meshing generally referred to as k-space equal to 0.1. The cohesive energy and lattice constant for all the materials were found to be in reasonable agreement with previous findings [62–65], thereby validating the initial methodology.

To examine the shear stress response, an orthogonal simulation cell with the X and Y axes aligned to the slip direction and slip plane normal respectively was employed. The simulation cell was deformed by applying incremental simple shear strain (γ_{xy}) along the active slip system till yielding was observed. The atomic positions and cell volume were fully relaxed while maintaining the applied shear strain. The ionic relaxations were based on the conjugate gradient method with energy and force convergence criteria of $1e^{-5}$ and 0.01 eV/Å, respectively. The shear stress response along the {111} <112> slip system was examined for Al and Ni. In the case of the body-centered cubic (BCC) metals (Fe and Nb), the {110} <111> slip system was selected, since at low temperatures this is the dominant slip system [66]. Lastly, the shear stress response along the <a> type prismatic slip system {10-10} <11–20> was chosen for Ti and Zr, as it has been found to be the dominant slip system at low temperature [67,68].

The effect of interstitial hydrogen on the ideal shear strength of different metals was quantified by varying interstitial hydrogen concentration of 2.7–10 at. % for all the metals, except Zr. In the case of Zr, the hydrogen concentration was varied from 1.33 to 5.88 at. % to avoid the hydrogen-hydrogen interactions. The simulation cell of $2 \times 1 \times 3$ periodic lengths along the X, Y, and Z axes, respectively was utilized for the cubic metals. For Ti, the simulation cell was built using $3 \times 3 \times 1$ periodicity along the X, Y, and Z axes, respectively. For Zr, $2 \times 2 \times 2$ periodicity was utilized for the simulation cell to achieve 3.03 and 5.88 at. % hydrogen concentration and $3 \times 3 \times 2$ periodicity was employed for 1.37 at. % hydrogen. Across all the metals, the simulation dimensions were chosen to minimize self-interactions between the hydrogen atoms. For instance, the solution energy for interstitial hydrogen (Equation (4)) converged with respect to different simulation cell dimensions. Furthermore, the calculated solid solution energy (Equation (4)) for interstitial hydrogen in different metals was found to be in good agreement with previous experimental and computational findings [69–76] (refer to Table S1).

$$E_s = E(M+H) - E(M) - \frac{1}{2}E(H_2) \tag{4}$$

Next, the partial density of states, Bader analysis, electron localization function, and valence charge difference distribution were examined to understand the effect of hydrogen on the electronic behavior of the host metal. The VESTA software tool was utilized to extract the planar charge density and electron localization function (ELF) contours [77].

Results

Ideal shear strength

In this section, the effect of hydrogen on the ideal shear strength of various metals will be discussed. The ideal shear strength of 3.09, 5.25, 8.07, 7.22, 2.61, and 1.84 GPa along the corresponding dominant slip system in Al, Ni, Fe, Nb, Ti, and Zr, respectively were obtained using first-principles calculations (Fig. 1), which is in good agreement with previously reported values [78–82], (refer to Table S3). To assess the effect of hydrogen on the shear response, a varying amount of interstitial hydrogen atoms were introduced to yield a wide range of hydrogen concentration (2.7–10.0 at. %). Next, to determine the equilibrium position of the hydrogen atom within the host lattice, the solution energy for interstitial hydrogen was quantified at the octahedral and tetrahedral sites (Table S2). Based on the minimum solution energy, it was found that hydrogen preferred the tetrahedral interstitial site in Al, Fe, Nb, and Zr (Table S2). On the other hand, in Ni and Ti, hydrogen preferred octahedral interstitial sites (Table S2). The increasing hydrogen concentration in Al and Ni led to a systematic decrease in the ideal shear strength (Fig. 1a,b). Furthermore, the presence of a 10 at. % hydrogen concentration led to a significant decrease in the ideal shear strength of Al ($\Delta\tau = 33\%$ at 10 at. % H) and Ni ($\Delta\tau = 12\%$@ 10 at. % H) when compared to the pure Al and Ni (3.09 and 5.27 GPa) (Fig. 1a,b). A similar systematic decrease in the ideal shear strength was observed with increasing hydrogen concentration for Fe ($\Delta\tau = 24.7\%$@ 10 at. % H) and Nb ($\Delta\tau = 10.9\%$@ 10 at. % H) (Fig. 1c,d). There was a similar effect of hydrogen on the ideal shear strength response of Ti and Zr (Fig. 1e,f). Notably, in Zr, the presence of interstitial hydrogen resulted in the biggest drop in the ideal shear strength ($\Delta\tau = 32.6\%$@ 5.88 at. % H, as

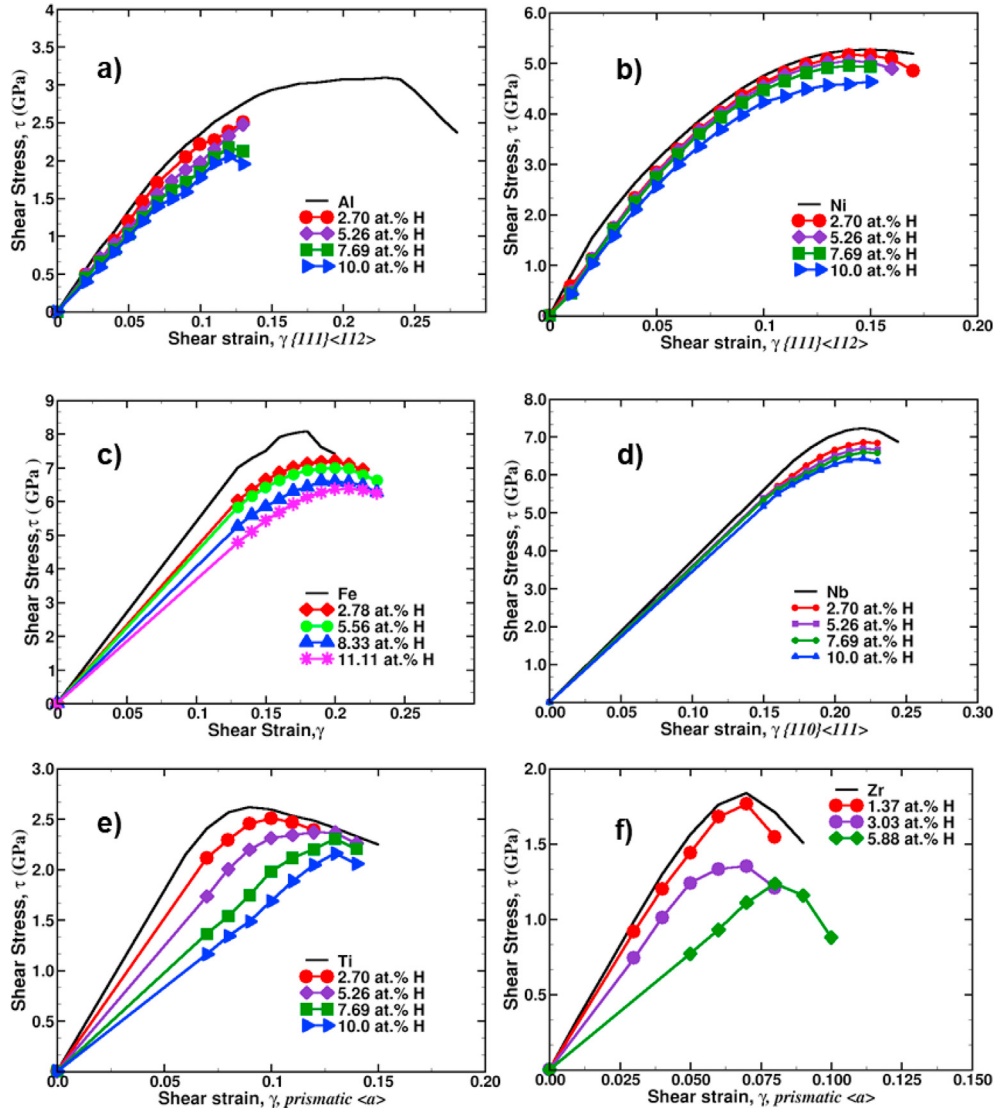

Fig. 1 – The effect of hydrogen on the shear stress-strain response of a) Al; b) Ni; c) Fe; d) Nb; e) Ti; and f) Zr along the respective dominant slip system.

compared to pure Zr) across the various metals examined here (Fig. 1).

Electronic insights

The presence of a foreign atom in the metal lattice noticeably alters the local atomic bonding, thus the examination of the electronic density of states (DOS) was utilized to understand the effect of hydrogen across the various metals. The change in the partial DOS (PDOS) of the metal atoms in the vicinity of the hydrogen atom was assessed to understand that the bonding between the valence electrons of the host and hydrogen atoms (Fig. 2). In pure Al, the PDOS for the valence 3*s* and 3*p* orbitals was uniformly spread across all the energy levels, unlike the sharp DOS curve of *d* orbital metals (Fig. 2a). The presence of hydrogen atom caused a small change in the PDOS of the Al atom in its immediate vicinity, suggesting a weak electronic interaction between the Al and interstitial hydrogen atoms. The rest of the metals examined here (Ni, Fe, Nb, Ti, and Zr) also have a sharp PDOS curve owing to the valence *d* orbital (Fig. 2b–f). The addition of interstitial hydrogen in *d* orbital metals resulted in a sharp peak at the energy level corresponding to the 1*s* orbital of H indicating a strong electronic interaction. Furthermore, the absence of a secondary peak (or peak widening with multiple peaks) around the energy level corresponding to the 1*s* orbital of H clearly shows an absence of H–H interactions within the host lattice.

To further gain insights into the local changes in electronic interactions due to the addition of interstitial hydrogen, a Bader charge analysis was carried out [83]. The Bader analysis approximately partitions the charge density between atoms and assigns it to individual atoms, thus providing a quantitative sense of charge redistribution between two atomic configurations. For all the host metals majority charge redistribution was observed for the first nearest neighbor atoms

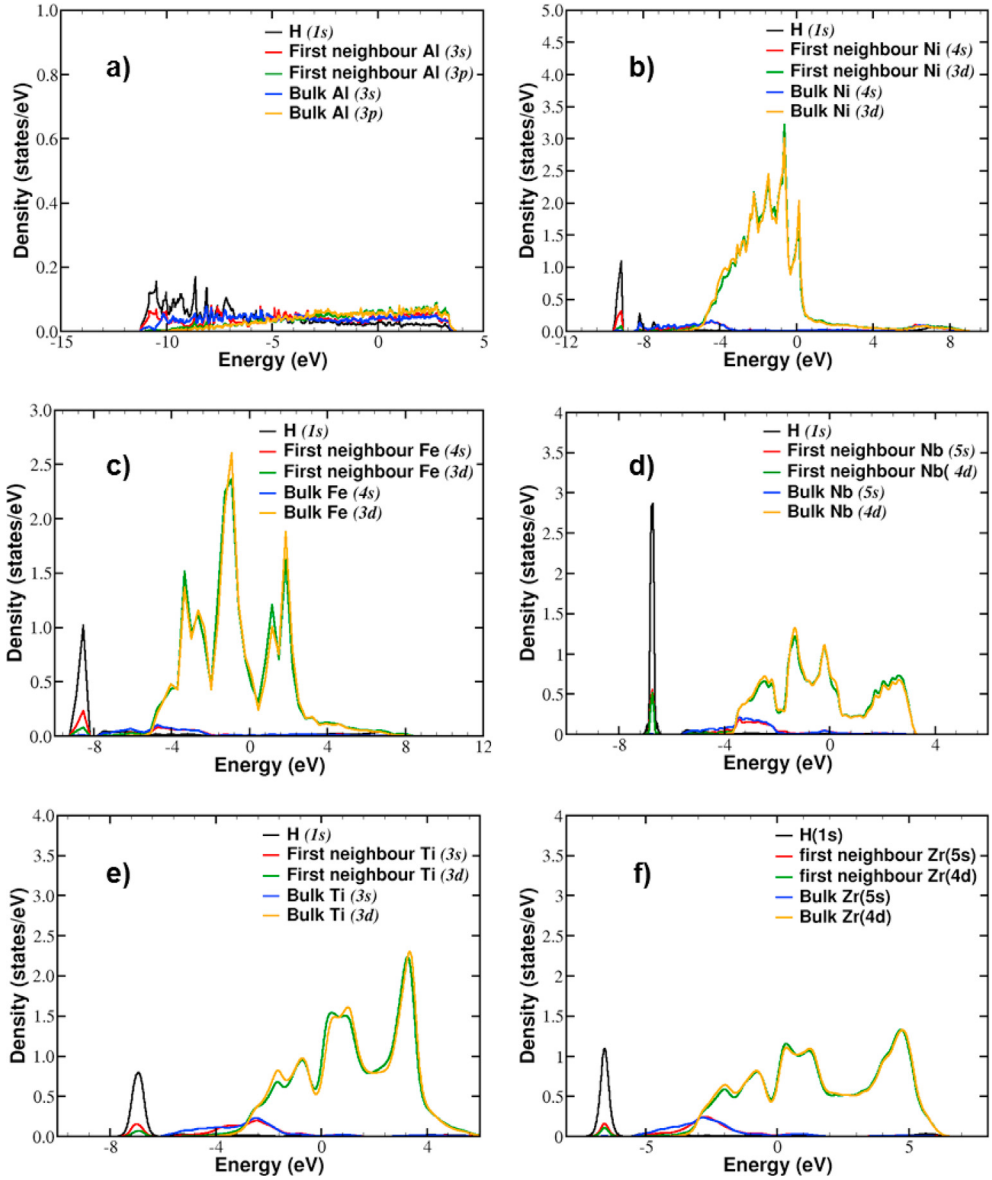

Fig. 2 – The effect of an interstitial hydrogen atom on the partial density of states of the first neighbor atom in a) Al; b) Ni; c) Fe; d) Nb; e) Ti; and f) Zr.

Table 1 – The net Bader charge on neighboring host metal and interstitial hydrogen atoms.

Material	Net Bader charge on H	Net Bader on the first neighbor host atom (Max/min)
Al	−1	+0.255/+0.245
Ni	−0.2	+0.03/+0.02
Fe	−0.39	+0.06/+0054
Nb	−0.62	+0.14/+0.11
Ti	−0.65	+0.06/+0.04
Zr	−0.76	+0.14/+0.13

due to the presence of interstitial hydrogen (Table 1), implying that hydrogen atom has localized bonding in all the host metal systems. In the case of Ni, the net Bader contribution of the first nearest Ni atoms towards hydrogen was found to be significantly less, when compared to other metals (Table 1). Interestingly in the case of Zr, the presence of interstitial hydrogen resulted in a small charge redistribution around the second neighbor Zr atom (Table 1).

The nature of bonding in the vicinity of hydrogen atoms can be further assessed via ELF, which quantifies the probability of finding two electrons of the same spin in space [84]. The ELF ranges between 0 and 1, where ELF around 0.5 indicates an ideal homogenous electron gas. Similarly, ELF >0.65 suggests a highly localized bonding (covalent if charge accumulation exists between atoms). To complement the ELF contours, the charge density difference was visualized to estimate the charge depletion and accumulation region after an interstitial atom was introduced. The charge density difference was calculated in the following manner:

$$\rho_{diff} = \rho(M+H) - \rho(M) - \rho(H) \quad (5)$$

where, $\rho(M)$, $\rho(H)$ and $\rho(M+H)$ were the charge density contributions of the pure metal atoms, a hydrogen atom, and a host metal with an interstitial hydrogen atom for the same simulation box, respectively. The addition of hydrogen in Al was found to cause a spherical accumulation of electrons around the hydrogen atom, a similar behavior was observed from the charge density contours (Fig. 3a,b). In the case of Ni and Fe, the electrons were delocalized (ELF in range of 0.1–0.2) and the presence of hydrogen had a negligible effect (Fig. 3c–f). In the case of Nb, the ELF was around 0.5 and the presence of interstitial hydrogen created a small degree of electron localization by causing a depletion of electrons from the neighboring Nb atoms (Fig. 3g–h). Interestingly in the case of pure Ti and Zr, there was a degree of electron localization near the interstitial sites (ELF around 0.65), suggesting a weak covalent tendency even for pure metals (Fig. 3i-l). The presence of interstitial hydrogen atoms to Ti and Zr leads to an increase in the extent of electron localization around the hydrogen atom (Fig. 3i-l). This electronic behavior was further examined using the charge density difference contours in Ti

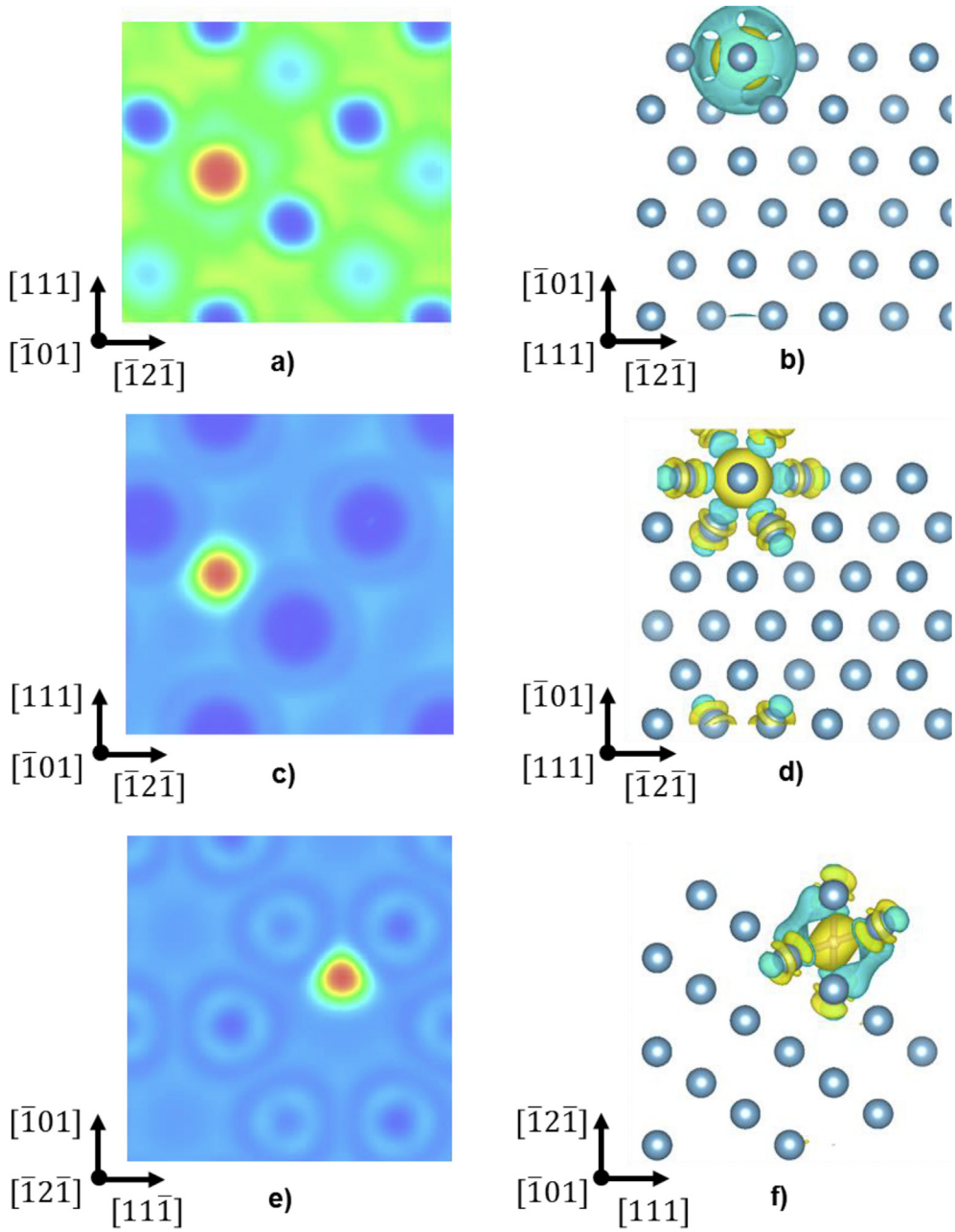

Fig. 3 – The electron localization function (left column) and charge density difference (right column) plots for (a–b) Al; (c–d) Ni; (e–f) Fe; (g–h) Nb; (i–j) Ti; and (k–l) Zr. The ELF contour plots were viewed along the crystallographic direction orthogonal to the respective slip systems. The charge density plots were viewed normal to the respective slip systems and the cyan and yellow iso surfaces correspond to +0.01 eV/A^3 and -0.01 eV/A^3 respectively across all the metals. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

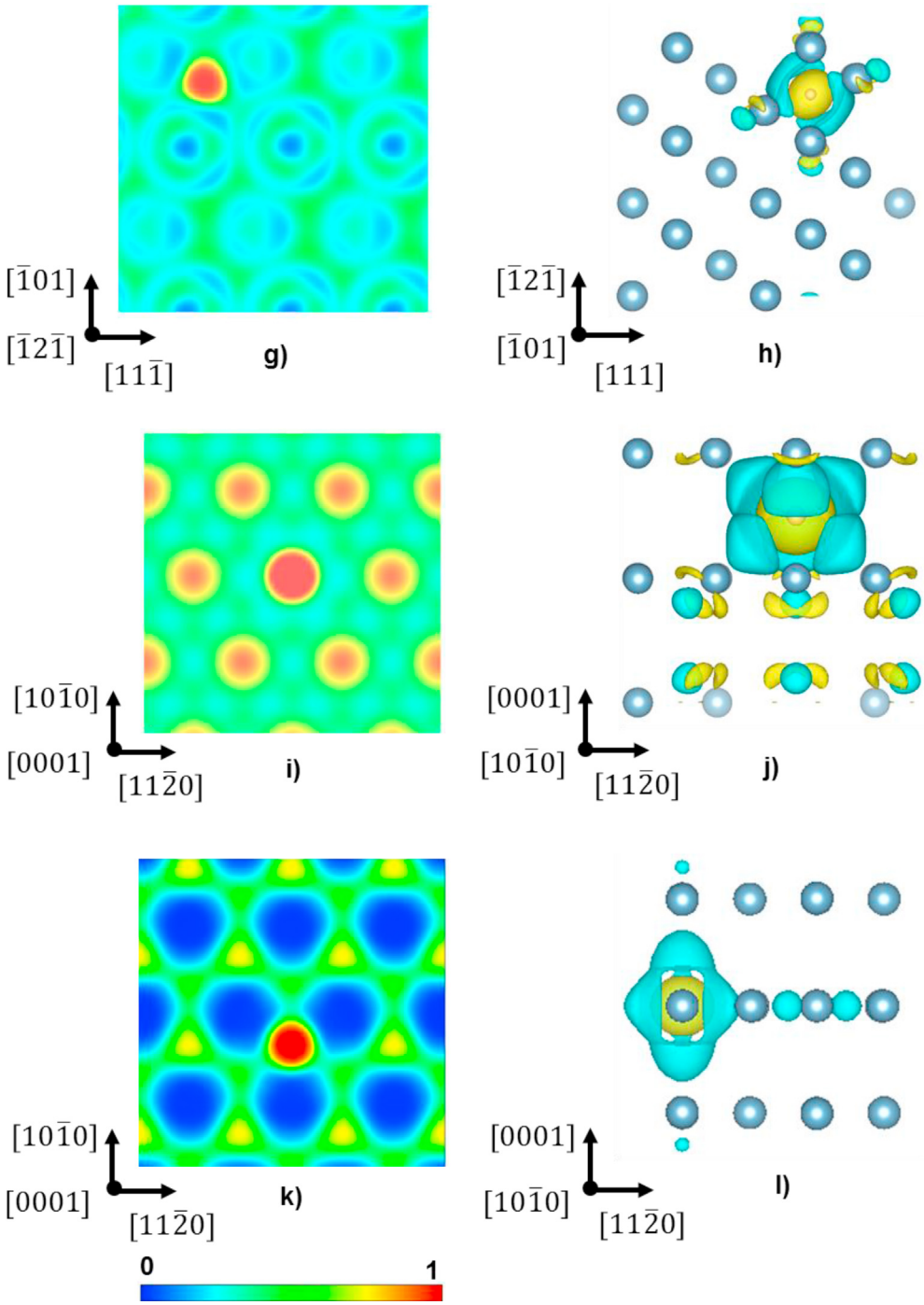

Fig. 3 – (*continued*).

and Zr (Fig. 3j and l). In the case of Ti, it was found that the charge depletion was hexagonally symmetric from the neighboring Ti atoms towards the interstitial hydrogen atoms (Fig. 3j). On the other hand, in Zr, the addition of interstitial hydrogen atoms caused anisotropic continuous charge depletion (Fig. 3l). Overall, the host metals that demonstrated a continuous region of charge depletion (i.e., Al and Zr, Fig. 3b and l), due to the addition of interstitial hydrogen atoms observed a larger drop in ideal shear strength, when compared to discrete regions of charge depletion (i.e., Ni and Fe, Fig. 3d and f).

Chemical-mechanical separation

The effects of interstitial hydrogen on the ideal shear strength of various metals can be further understood by quantifying the chemical and mechanical contributions [56,85,86]. The chemical contribution accounts for the alterations in the

charge transfer and local bonding behavior that take place due to the presence of interstitial hydrogen atoms (τ_{IS}^{chem}). On the other hand, the addition of interstitial hydrogen atoms can cause the volumetric expansion of the host lattice. The effect of volumetric distortion on the ideal shear strength is captured by the mechanical contribution (τ_{IS}^{mech}). Therefore, the ideal shear strength of the host metal for a given hydrogen concentration (τ_{IS}) can be expressed in the following manner:

$$\tau_{IS} = \tau_{IS}^{mech} + \tau_{IS}^{chem} \quad (6)$$

The mechanical contribution was determined by removing the interstitial hydrogen atoms from a fully relaxed metal-hydrogen supercell and the atoms were allowed to adjust their positions, while the supercell volume was held fixed. This was repeated across the wide range of applied shear strain and hydrogen concentration for the different metals examined here. On the other hand, the chemical contribution was found by introducing interstitial hydrogen atoms into a fully relaxed metal supercell, while the atomic positions and supercell volume were held fixed. However, once the mechanical contribution was found, the chemical contribution can be deduced with the help of Equation (6). The addition of interstitial hydrogen led to a linear increase of the volumetric strain experienced by the various metals examined here (Fig. 4a). The addition of interstitial hydrogen created a large volumetric strain in the cubic metals, especially Fe and Al. On the other hand in HCP metals, the addition of interstitial hydrogen led to a smaller volumetric expansion in the comparatively larger crystal lattices.

Figure 4b shows the $\tau_{IS}^{mech}/\tau_{IS}$ ratio across a wide range of hydrogen concentrations for the various metals. Overall, the mechanical contribution (volumetric distortion) due to the addition of interstitial hydrogen had a significant effect on the ideal shear strength. In the case of Ti and Zr, the $\tau_{IS}^{mech}/\tau_{IS}$ the ratio was found to be greater than 1 across the wide range of hydrogen concentration (Fig. 4a). This trend implies that the chemical contribution had a significant influence on the ideal shear strength of Ti and Zr (Fig. 4a). This observation can be understood with the help of ELF and charge density difference plots for Ti and Zr discussed earlier (Fig. 3i-l), as both Ti and Zr show a large change in charge density and ELF due to the addition of interstitial hydrogen. In the case of Ni, Fe, and Nb, the $\tau_{IS}^{mech}/\tau_{IS}$ the ratio was near 1 across a wide range of hydrogen concentrations, suggesting that the chemical contribution was significantly smaller and counteracting (negative magnitude) the mechanical contribution. Furthermore, the difference in the effect of interstitial hydrogen on ideal shear strength in Fe and Nb can be attributed to the larger volumetric strain developed in Fe leading to the increased mechanical contribution, as compared to Nb (Fig. 4b). On the other hand, the chemical contribution to the ideal shear strength of Al was positive, as interstitial hydrogen atoms do not prefer the Al lattice.

Discussion

So far, using several first-principles calculations, the effect of interstitial hydrogen on the ideal shear strength (τ_{IS}) across different metals (Al, Ni, Fe, Nb, Ti, and Zr) was quantified. The lattice resistance to shear deformation was found to decrease across all the metals examined here, primarily due to the volumetric strain introduced by interstitial hydrogen atoms. Based on the DOS analysis, it was found that the metals with valence *d* orbital (Ni, Fe, Nb, Ti, and Zr) demonstrated strong interaction with hydrogen. In the case of Ni and Fe, the electrons were delocalized, but for Ti and Zr there was a strong tendency for localization. Furthermore, the extent of charge redistribution due to the presence of hydrogen in Ti and Zr had a crucial role in understanding the large decrease in ideal shear strength with increasing hydrogen concentration. The role of chemical interactions in Ti and Zr was also validated by quantitatively separating the chemical and mechanical contributions of interstitial hydrogen on the ideal shear strength. To provide a broader perspective and to reconcile some of the previous experimental findings, the implications of the observed variation in ideal shear strength due to the presence of hydrogen on the macroscopic plasticity were examined by qualitatively assessing the Peierls stress.

The analytical expression developed by Joos and Duesbery [58] to estimate the Peierls stress which is based on the Peierls Nabarro (PN) formulation [87–89] was employed (refer to Equation (1) and Table S3). In this analysis, we assume that both Burger's vector (b) and atomic spacing (a') remain

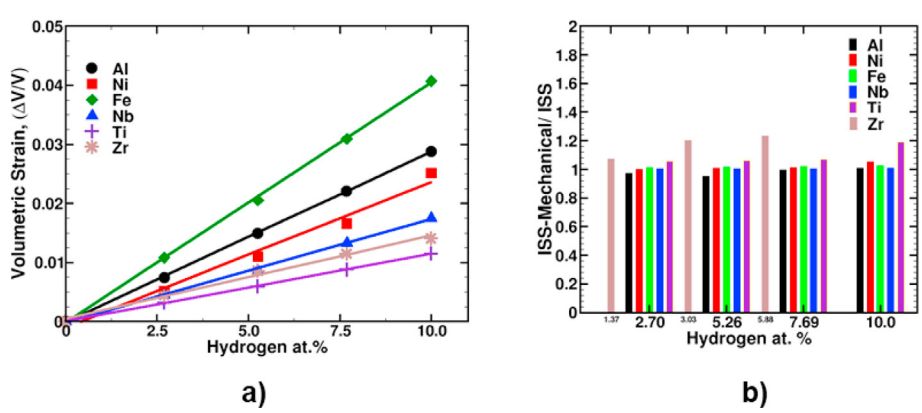

Fig. 4 – a) The variation of the mechanical contribution of ideal shear strength normalized with ideal shear strength with the atomic percentage of hydrogen and b) change in volumetric strain of host lattice with the addition of an atomic percentage of hydrogen.

unchanged due to the addition of interstitial hydrogen. Next, the isotropic shear modulus (μ) of the host metal was approximated by the Voigt shear modulus average for isotropic solid as given in Equation (6).

$$15\mu = (C_{11} + C_{22} + C_{33}) - (C_{12} + C_{23} + C_{31}) + 3(C_{44} + C_{55} + C_{66}) \quad (7)$$

The values of various elastic constants (C_{ij}) for pure metals were taken from previous DFT findings [90–94]. The ideal shear strength of a crystalline solid is intrinsically related to the shear modulus, thus we assumed that the ratio between the shear modulus and ideal shear strength remains constants across varying hydrogen concentrations in different metals. This enables a qualitative estimation of hydrogen on the Peierls stress of a screw dislocation across various metals (Fig. 5), for details refer to Supplementary Table S3.

The addition of interstitial hydrogen led to a decrease in Peierls stress across the metals examined here, similar to the ideal shear strength findings (Figs. 5 and 1). However, in the case of Zr, increasing the hydrogen concentration resulted in a significant drop in the Peierls stress. This trend of Peierls stress suggests that interstitial hydrogen is a key defactant that reduces the energy for dislocation nucleation and motion [42,95]. This analysis attempts to capture the effect of interstitial hydrogen in the bulk metal, thus the competing hardening observed when hydrogen segregates at the dislocation core is not explained here.

The observed reduction in Peierls stress, can in turn, lead to increased dislocation velocity and slips localization, as proposed by the defactant, AIDE, and HELP mechanisms. The HELP mechanism is based on the idea that the elastic shielding provided by hydrogen atoms segregated around the dislocation results in a decrease in the repulsive interactions between dislocations, which finally yields enhanced dislocation mobility [5,6,36–39]. However, the decrease in lattice friction to shear deformation across various metals observed in this work does not explain the HELP mechanism. Interestingly, the adsorption of external hydrogen at stress concentration sites drives the inelastic response in the AIDE mechanism [40]. The defactant model proposed by Kirchheim [41,42] suggested that hydrogen segregates to pre-existing defects and reduces the stress required for homogenous dislocation nucleation, which has been supported by various experiments observations. The present findings clearly found that interstitial hydrogen acts as a defactant and led to a decrease in the stress required for dislocation nucleation, thereby creating the microscopic conditions for HELP/AIDE and HEDE mechanisms to act at the later stages of deformation to result in overall HE.

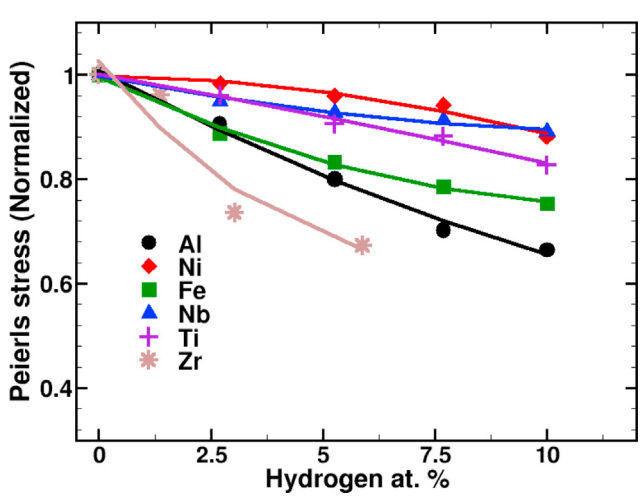

Fig. 5 – The variation of normalized Peierls stress at different hydrogen percentages relative to the Peierls stress of pure material (quadratic fitting to data points).

Conclusions

A comprehensive first-principles study was carried out to understand the implications of interstitial hydrogen atoms on the plastic deformation of various metals (Al, Ni, Fe, Nb, Ti, and Zr). Specifically, the effect of hydrogen concentration on the ideal shear strength along the dominant slip systems of various metals was examined. Furthermore, the electronic structure (DOS), Bader analysis, nature of bonding (charge density and ELF contours), chemical-mechanical separation, and Peierls stress estimation was carried out to uncover the reasons for the observed shear response across the different metals. The following were the key findings of this work:

1. The presence of increasing hydrogen concentration led to a continuous decrease in the ideal shear strength along the dominant slip system across all the metals examined here. However, it was found that the lattice resistance to shear deformation in Ni and Nb were least susceptible to interstitial hydrogen.
2. Based on electronic insights, we found that interstitial hydrogen atoms had short-range interactions confined to the nearest neighbor metal atoms. For instance, the Bader analysis found that the amount of charge accumulation that took place due in the vicinity of the hydrogen atom was provided by the nearest neighbor host atoms (Table 1). Furthermore, the charge difference and ELF contours also support these findings (Fig. 2).
3. In the case of Al, it was found that the electron distribution was homogenous even before the addition of hydrogen. Furthermore, there was a continuous spherical charge density depletion from the neighboring Al atoms. However, the interstitial sites in Al lattice are highly unfavorable sites for hydrogen atoms. This behavior also explains the observation that both the chemical and mechanical contributions lead to a decrease in ideal shear strength of Al with increasing hydrogen concentration. Nonetheless, the volumetric strain developed, due to the addition of hydrogen atom in Al lattice was the major contributing factor in the decrease of ideal shear strength.
4. In the case of HCP metals, the addition of hydrogen atoms led to the smallest volumetric strain (Fig. 3b). Nonetheless, the mechanical contribution had a significant role in the observed decrease of ideal shear strength with increasing hydrogen concentration. However, the chemical contribution in HCP metals was noticeable and counteracting the mechanical contribution towards ideal shear strength, as compared to BCC and FCC metals (Fig. 3a).
5. Based on PN formulation, the lattice resistance for screw dislocation was found to decrease with increasing hydrogen concentration. These findings support the notion

that the presence of hydrogen results in increased dislocation density and velocity. Finally, suggesting that the increased plastic deformation at the macroscopic level plays a crucial role in the overall hydrogen embrittlement mechanism, depending on the applied strain rate, temperature, and hydrogen concentration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijhydene.2021.05.068.

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